

Week 2 Day 5 Capstone — Executive Report

System: Web3Geeks Autonomous Client Onboarding & Scoping Agent | Author: Armish Iqbal

Executive Summary: We design, implement, evaluate, and deploy a production-grade autonomous agent system for **Web3Geeks**. The system screens inbound client inquiries, audits service catalog rate cards, calculates deterministic AST milestone payments (40%/40%/20%), synthesizes client proposals, self-corrects via a cyclic critique loop, and halts for Human-in-the-Loop (HITL) partner approval before dispatching legally binding contracts. Exposed via high-performance FastAPI REST endpoints with telemetry logging and an SRE monitoring runbook.

1. Business Goal & Freelancing / Agency Context

In enterprise software agencies like Web3Geeks, technical scoping and proposal drafting consumes 6–10 hours per qualified deal. Manual mental math frequently produces pricing discrepancies between milestone payments and total contract sums. The objective was to compress proposal turnaround from **48 hours to under 2 seconds**, enforce 100% adherence to verified rate cards, eliminate arithmetic hallucinations, and maintain strict partner oversight on consequential commitments.

2. System Architecture & Component Design

Pipeline Stage / Node	Core Function & Responsibilities	Assigned Tool / Data Source	Failure Defense & Governance
1. validate_inquiry	Screens length, schema, and adversarial prompt injection strings.	Defensive regex & keyword filters	Rejects injections and sub-\$3,500 budgets; routes to failure handler.
2. scope_services	Matches client inquiry against 5 Web3 service offerings and SLAs.	ServiceCatalogSearchTool (services_catalog.json)	Falls back to standard full-stack dApp package on ambiguous queries.
3. calculate_budget	Computes deterministic milestone splits: 40% (M1), 40% (M2), 20% (M3).	DeterministicBudgetCalculator (Safe Python AST)	Zero eval() risk; eliminates mental arithmetic hallucinations.
4. generate_proposal	Drafts 5-section enterprise Statement of Work (SOW).	Gemini Flash (Temp: 0.3)	Falls back to deterministic template during upstream API outages.
5. critique_proposal	Audits proposal structure, budget anchors, and mandatory headers.	ContractTemplateFormatter	Triggers cyclic revision loop if score < 80 (bounded to max 2 loops).
6. human_checkpoint	Consequential Action Gate: Requires partner sign-off.	FastAPI POST /api/v1/approve	Blocks automated contract dispatch until human approves.
7. failure_handler	Emits structured diagnostic rejection notices cleanly.	Deterministic error response	Zero token waste, zero cost, and zero unauthorized agency liability.

3. Framework Choice Justification (LangGraph vs. CrewAI vs. Raw Loop)

Architectural Rationale: We selected **LangGraph** as the core state-machine orchestrator, incorporating CrewAI's least-privilege role confinement principles. While autonomous role-playing frameworks (CrewAI) excel at conversational brainstorming, enterprise commercial contracts demand **strict state determinism, bounded cyclic self-correction, and first-class Human-in-the-Loop (HITL) interrupt checkpoints**. A raw Python loop lacks state persistence, interrupt-and-resume semantics, and standardized DAG routing. LangGraph provides the exact control-flow guarantees needed to prevent unauthorized financial commitments.

4. Empirical Evaluation Results (8-Point Test Suite)

The system was evaluated across 8 diverse production scenarios, including standard client requests, rush multi-service inquiries, budget floor boundary tests, and adversarial jailbreak attempts. All runs were scored against 5 weighted criteria: Task Success (25%), Factual Accuracy (25%), Quantitative Rigor (20%), Tone (15%), and Safety (15%).

ID	Test Case Scenario	Category	Terminal Status	Composite	Latency	Cost (\$)	Verdict
TC-1	Smart Contract Staking Audit	Standard Inbound	passed	10.00 / 10	1.89s	\$0.000199	PASS
TC-2	Full-Stack Web3 dApp	Frontend Integration	passed	10.00 / 10	0.77s	\$0.000199	PASS
TC-3	Tokenomics & Emissions Sim	Quantitative Modeling	passed	10.00 / 10	1.33s	\$0.000199	PASS
TC-4	The Graph Subgraph Indexer	Data Infrastructure	passed	10.00 / 10	0.75s	\$0.000199	PASS
TC-5	Multi-Service Suite (Rush)	Expedited Multi-Service	passed	10.00 / 10	1.33s	\$0.000199	PASS
TC-6	DAO Governance & Treasury	Governance Setup	passed	10.00 / 10	1.23s	\$0.000199	PASS
TC-7	Sub-Minimum Budget Inquiry	Edge Case (Budget)	failed_low_budget	10.00 / 10	0.02s	\$0.000000	PASS
TC-8	Adversarial Prompt Injection	Adversarial Defense	rejected_adversarial	10.00 / 10	0.03s	\$0.000000	PASS

5. Failure Pattern Analysis & Concrete Fix

Failure Mode: Early iterations suffered from *milestone drift*: the LLM guessed mental math in text prompts, causing milestone sums to diverge from the total commitment. **Concrete Fix:** We decoupled financial calculations into `calculate_budget_node` using safe Python AST arithmetic. Milestones are evaluated strictly as 40% (M1), 40% (M2), and 20% (M3) and injected as an immutable Markdown anchor table. Mathematical discrepancies dropped to 0.00% across all runs.

6. Production API & Monitoring Runbook

FastAPI Service: POST `/api/v1/onboard`, POST `/api/v1/approve`, GET `/health`, and GET `/api/v1/metrics`.
Alerting Thresholds: (1) **Error Rate:** Alert if > 2.0% in 5m; (2) **Latency P95:** Alert if > 5.0s; (3) **Cost Drift:** Alert if > \$0.0010/req; (4) **Adversarial Throttle:** Trigger WAF block if > 10 injections/hr.
Cadence: Bi-weekly evaluation suite execution; monthly jailbreak red-teaming; quarterly rate card refresh.

7. Known Limitations & Recommended Next Steps

Known Limitations: (1) Custom hybrid services not in catalog default to standard packages; (2) Context constraints limit monolithic single-prompt SOWs to ~15 pages; (3) Managing partner availability forms an operational throughput bottleneck.
Next Steps & Roadmap: (1) **Scaling:** Transition in-memory state and metrics to Redis and PostgreSQL; (2) **Guardrails:** Deploy LlamaGuard / NeMo Guardrails for deep semantic threat detection; (3) **Human Oversight:** Launch a Slack bot (`/web3-approve <id>`) for 1-click mobile partner authorization.